



UNIVERSITI UTARA MALAYSIA
SECOND SEMESTER SESSION 2025/2026 (A252)

STISK3113 DATA VISUALIZATION

GROUP PROJECT

PREPARED FOR:

Dr. Nik Fatinah binti N. Mohd Farid

PREPARED BY: GROUP 2

NAME	MATRIC NO
1) Shiva Shangkar Murugan	301907
2) Ng SheyRu	300210
3) Kok Xiao Wei	299935
4) Seet Zhen Ern	299727
5) Nishkalla A/P Selva Ilangowan	299980

1. Introduction

Data visualization plays an important role in transforming raw healthcare data into meaningful information that supports decision-making. In healthcare settings, understanding patient flow, waiting times, and patient satisfaction is essential for improving service quality and operational efficiency.

This project utilizes the RFID Timestamp and Electronic Survey Data for Patient Experience dataset obtained from Mendeley Data. The objective of this project is to analyze outpatient clinic processes and patient satisfaction levels through interactive visualizations. A dashboard was developed using Power BI to present key insights related to patient visit duration, waiting times, operational performance, and patient satisfaction.

2. Dataset Description

The project uses two datasets:

2.1 RFID Timestamp Dataset

The RFID dataset records the movement of patients through several service stations in an outpatient clinic. Each record contains timestamps for:

- Registration arrival, process start, and process end
- Vitals arrival, process start, and process end
- Doctor station arrival, process start, and process end

From these timestamps, several performance indicators were calculated, including:

- Registration waiting time
- Vitals waiting time
- Doctor waiting time
- Total patient visit time

The RFID dataset consists of records collected between May 2019 and November 2021.

2.2 Electronic Survey Dataset

The survey dataset contains patient responses to 18 satisfaction-related questions (q1–q18) and an Overall Satisfaction Index (OSI).

The responses are measured using a Likert-scale rating system. The survey evaluates different aspects of patient experience, including service quality, waiting time, staff interaction, and overall satisfaction.

The OSI variable was used as the primary indicator of patient satisfaction throughout the analysis.

3. Interaction and Task Categories

Several interaction and task categories were implemented in the dashboard to support exploratory analysis.

3.1 Interaction Categories

The dashboard includes the following interaction features:

Filtering

Users can filter the dashboard based on:

- Year
- Month
- Day
- OSI category
- Waiting time categories

Drill-Down Analysis

Decomposition tree visualizations allow users to drill down from:

- Year → Month → Day
- OSI → Waiting Times → Total Visit Time

Highlighting and Cross-Filtering

Selecting data points in one visualization automatically updates related visualizations on the dashboard.

3.2 Task Categories

The dashboard supports several analytical tasks:

Trend Analysis

- Average total visit time by month
- Patient volume trend over time

Comparison

- Comparison of waiting times across different service stages
- Comparison of satisfaction levels by OSI category

Relationship Analysis

- Relationship between total visit time and patient satisfaction
- Relationship between doctor waiting time and OSI

Distribution Analysis

- Distribution of patients by arrival hour
- Distribution of patients across OSI categories

Flow and Network Analysis

- Patient arrival hour to waiting category flow using a Sankey diagram

Hierarchical Analysis

- Decomposition tree analysis of visit time and satisfaction metrics

4. Task Description and Key Findings

The dashboard was developed to investigate outpatient operational performance and patient satisfaction.

Patient Visit Duration

The analysis shows that the average patient visit duration varies across different months. Higher average visit times were observed during certain periods, indicating potential congestion and increased waiting times within the clinic.

Patient Volume Trends

Patient volume increased significantly from May to September before showing a slight decline in later months. This trend may reflect fluctuations in healthcare demand throughout the observation period.

Waiting Time Performance

Doctor waiting time contributed noticeably to the overall patient visit duration. The average doctor waiting time was approximately 8.73 minutes, suggesting that delays at the consultation stage influence overall service efficiency.

Patient Satisfaction

The Overall Satisfaction Index (OSI) showed an average value of approximately 4.04, indicating generally positive patient perceptions of the healthcare services provided.

Relationship Between Visit Duration and Satisfaction

The scatter plot analysis suggests that patient satisfaction remained relatively stable across different visit durations. While some patients experienced longer visits, satisfaction levels generally remained within the moderate to high range.

Arrival Hour Analysis

The arrival-hour visualization revealed that patient arrivals were concentrated during specific morning and afternoon periods. These peak periods may contribute to increased waiting times and resource utilization.

Flow Analysis

The Sankey diagram demonstrated the relationship between patient arrival hours and waiting categories. Most patients were classified within the short waiting-time category, indicating relatively efficient patient processing throughout the clinic.

5. Limitations

Several limitations were identified during the project.

1. The dataset does not contain textual comments or open-ended feedback from patients. Therefore, advanced text visualization techniques such as word clouds, sentiment analysis, and text mining could not be performed.
2. The survey dataset only contains numerical ratings, limiting the ability to analyze detailed patient opinions and experiences.
3. The RFID and survey datasets were analyzed primarily through aggregated measures, which may not capture all individual patient experiences.
4. The study is limited to the available data collection period and may not represent long-term trends in outpatient services.

6. Conclusion

This project successfully applied data visualization techniques to analyze outpatient clinic operations and patient satisfaction using RFID timestamp data and electronic survey responses. The dashboard provided insights into patient flow, waiting times, visit duration, operational performance, and satisfaction levels.

The findings indicate that patient arrivals and waiting times vary across different periods, while overall patient satisfaction remains relatively positive. Interactive visualizations such as decomposition trees, scatter plots, line charts, Sankey diagrams, and KPI dashboards enabled effective exploration of healthcare performance metrics.

Overall, the dashboard demonstrates how data visualization can support healthcare organizations in monitoring operational efficiency and improving patient experience through data-driven decision-making.